

IPCC AR6 — Summary for Policymakers

A Learner's Guide to the Sixth Assessment Report on Climate Change

Reading 1 · Module 1, Topics 1.1–1.3 · Meer Foundation EduSuTo

Course: Climate Change: Carbon Capture and Storage

What Is the IPCC AR6?

The Intergovernmental Panel on Climate Change (IPCC) is the United Nations body for assessing the science related to climate change. Every 6–7 years, the IPCC releases an Assessment Report synthesising the latest peer-reviewed climate research from thousands of scientists worldwide. The Sixth Assessment Report (AR6) was released in stages between 2021 and 2023. This reading summarises the key findings most relevant to this course.

Key Finding 1: The Scientific Basis is Unequivocal

Human influence has warmed the climate at an unprecedented rate. Global surface temperature has increased by approximately 1.1°C above the 1850–1900 average — and this warming is unequivocally attributed to human activities, primarily the burning of fossil fuels and land-use change. The AR6 used stronger language than any previous IPCC report.

Key Statistic: Each of the last four decades has been successively warmer than any decade that preceded it since 1850.

Key Finding 2: Every Fraction of a Degree Matters

The difference between 1.5°C and 2°C of warming is not academic. AR6 quantifies the impacts at each level:

Impact Category	At 1.5°C Warming	At 2°C Warming
Sea level rise by 2100	~0.43 m	~0.54 m
Arctic summers ice-free	Once per 100 years	Once per 10 years
Species losing habitat (vertebrates)	4%	8%
People exposed to heat stress	~1.6 billion	~2.0 billion
Coral reef loss	70–90%	99%

Key Finding 3: Greenhouse Gas Concentrations and Sources

Atmospheric CO₂ concentration reached 419 ppm in 2023 — higher than at any point in at least 2 million years. The main contributors to the 2010–2019 increase in GHG emissions:

Sector	Share of Global GHG Emissions
Energy supply (electricity, heat)	34%
Industry (cement, steel, chemicals)	24%
Agriculture, forestry, land use	22%
Transport	15%
Buildings	6%

Key Finding 4: Carbon Budgets and the Remaining Window

The concept of a carbon budget is central to climate science and to this course. A carbon budget is the total cumulative amount of CO₂ that can be emitted while keeping warming below a specified temperature target.

Temperature Target	Probability	Remaining Budget (as of 2023)	Years at Current Emissions
1.5°C	50%	~250 GtCO ₂	~6 years
1.5°C	67%	~150 GtCO ₂	~4 years
2.0°C	50%	~1,150 GtCO ₂	~29 years
2.0°C	67%	~900 GtCO ₂	~22 years

Critical insight: The carbon budget for 1.5°C at 50% probability is equivalent to approximately 6 years of current global emissions (~40 GtCO₂/year). This makes deep emission cuts and large-scale carbon dioxide removal both urgent and non-negotiable.

Key Finding 5: Mitigation Pathways — What AR6 Says About CCS

The AR6 Working Group III report (Mitigation of Climate Change) analysed hundreds of scenarios for limiting warming to 1.5°C or 2°C. Its findings on Carbon Capture and Storage are clear:

- All 1.5°C-compatible scenarios in AR6 rely on CCS and Carbon Dioxide Removal (CDR) to some extent.
- The median 1.5°C scenario requires 6–16 billion tonnes of CDR per year by 2050, of which CCS-based removal (BECCS, DAC) contributes 2–8 GtCO₂/year.

- Industrial CCS (cement, steel, chemicals) is included in 90%+ of all 1.5°C-compatible scenarios as a necessary mitigation measure.
- The cost of inaction far exceeds the cost of mitigation. Every dollar invested now saves multiple dollars in future climate damages.

Key Finding 6: Tipping Points and Irreversibility

AR6 identifies several climate tipping points that could be crossed even below 2°C of warming. These are thresholds beyond which changes become self-reinforcing and effectively irreversible on human timescales:

Tipping Element	Threshold (°C above pre-industrial)	Consequence
West Antarctic Ice Sheet collapse	1.5–2°C	3–5 m sea level rise over centuries
Greenland Ice Sheet loss	1.5–3°C	7 m sea level rise over millennia
Amazon dieback	3–4°C	Loss of 50%+ of Amazon rainforest
Permafrost degradation	1.5–2°C	Massive CH ₄ and CO ₂ release
Atlantic circulation disruption (AMOC)	Uncertain	Major European cooling, monsoon disruption

India and the IPCC AR6: Key Findings

The AR6 includes extensive regional assessments. For South Asia and India, key findings include:

- India will experience increasingly severe and frequent extreme heat events, with 1.5°C of warming making heat waves that currently occur once per 50 years occur once per 5 years.
- Indian monsoon variability is projected to increase, with more intense rainfall events and more severe droughts between rain events.
- The Himalayan cryosphere is warming at twice the global average rate; glacial melt threatens water security for hundreds of millions.
- India's agricultural productivity faces declining yields for rice and wheat under 1.5–2°C scenarios without adaptation measures.
- India has a large mitigation potential — including solar, wind, energy efficiency, and nature-based solutions including soil carbon and agroforestry.

How to Read the Full AR6

The full AR6 comprises three Working Group reports (WGI: Physical Science Basis; WGII: Impacts, Adaptation and Vulnerability; WGIII: Mitigation) plus a Synthesis Report. For this course, the most relevant documents are:

Document	Relevance to This Course	Where to Find
WGI SPM	Climate science foundations	ipcc.ch/report/ar6/wg1
WGIII SPM	Mitigation pathways, CCS role	ipcc.ch/report/ar6/wg3
Synthesis Report SPM	Overall picture for policymakers	ipcc.ch/ar6/syr
WGIII Chapter 6	Energy systems and CCS	ipcc.ch/report/ar6/wg3
WGIII Chapter 7	Agriculture, forests, CDR	ipcc.ch/report/ar6/wg3